3 Set up the turtle

The code shown here calls functions in the `turtle` module to set the background colour, as well as the turtle's speed and size.

```
import turtle

turtle.bgcolor('black')
turtle.speed('fast')
turtle.pensize(4)
```

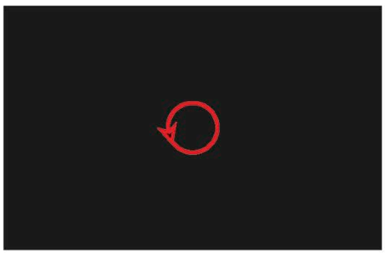
Background colour

The turtle's speed

The thickness of the turtle's trail

4 Choose the pen colour, draw a circle

Next set the colour of the turtle's trail and test the code by drawing a circle. Add these two lines to the end of your code and run the program.



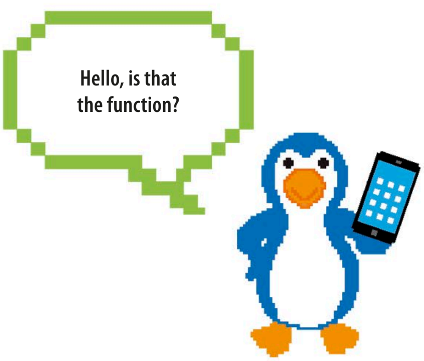
```
import turtle

turtle.bgcolor('black')
turtle.speed('fast')
turtle.pensize(4)

turtle.pencolor('red')
turtle.circle(30)
```

Pen colour

This tells the turtle to draw a circle.

5 Draw more circles

You should now see a single circle, but we need lots more. Here comes the clever bit. Put the commands to draw a red circle inside a function, but add a line so that the function calls itself. This trick, known as recursion, makes the function repeat. Remember, functions need to be defined before they're used, so you'll need to move the function above the line where it's called.




```
import turtle

def draw_circle(size):
    turtle.pencolor('red')
    turtle.circle(size)
    draw_circle(size)

turtle.bgcolor('black')
turtle.speed('fast')
turtle.pensize(4)
draw_circle(30)
```

This line uses the `size` parameter.

The function calls itself, which makes it repeat endlessly.

This line calls the function for the first time.